

21-127 Induction

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Prove that $(1+x)^{21,127} \geq 1 + 21,127 x$ for all real number $x \geq -1$.

Solution. We will prove that for all real $x \geq -1$ and any integer $n \geq 1$,

$$(1+x)^n \geq 1+nx$$

Let $x \in \mathbb{R}$ such that $x \geq -1$. We proceed by induction on $n \geq 1$.

- **Base Case:** Let $n = 1$. Then $(1+x)^1 = 1+x = 1+1 \cdot x$, so the inequality holds.
- **Inductive Case:** Let $n \in \mathbb{Z}$ such that $n \geq 1$.
- **Inductive Hypothesis:** Assume that $(1+x)^n \geq 1+nx$.
- **Inductive Step:** Consider $n+1$. Then

$$(1+x)^{n+1} = (1+x)^n(1+x).$$

Using the inductive hypothesis, we have

$$(1+x)^{n+1} \geq (1+nx)(1+x) = 1 + (n+1)x + nx^2.$$

Since $x^2 \geq 0$, it follows that

$$(1+x)^{n+1} \geq 1 + (n+1)x.$$

By the Principle of Mathematical Induction, we conclude that for all integers $n \geq 1$

$$(1+x)^n \geq 1+nx$$